PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONGIDO DISTRICT EXAMINATION SYNDICATE

FORM FOUR PRE-MOCK EXAMINATION

BASIC MATHEMATICS

041 **Monday, 22nd March 2021 morning** TIME: 3 HOURS

Instructions

1. This paper consists of section A and B with total of fourteen(14) questions
2. Answer all questions.
3. Show all of your working clearly
4. Mathematical devices and any non-programmable calculator may be used.

SECTION A (60 Marks)

1. (a) Express the number 39.996
   (i) To four significant figure
   (ii) To two decimal place
   (iii) To expanded form

   (b) In student leadership election $\frac{1}{3}$ of the students voted for candidates A, $\frac{8}{15}$ voted for candidates B, and the rest did not vote.

   i. What fraction voted? ii. What percentage did not vote.
2. (a) If $\frac{\sqrt{3}}{2+\sqrt{3}}$ = r+ M$\sqrt{P}$ find the value of r+ M+P

   (b) Determine the value of x if$\left(\frac{1}{1024}\right)^{x+3}=32^{x+3}$ (c) Find the value of x if 2logx =log64/log2
3. In class of 45 pupils ,19 study commerce but not physics, 16 study physics but not commerce and 3 studies neither commerce nor physics
   (i) how many study both commerce and physics?
   (ii) how many study physics?
   (b) A bag contain ball and 10 blue balls. If one ball is drawn from the bag without replaced and the second ball is drawing. Find the probability of drawing ball of different colour.
4. Find the equation of line which is perpendicular bisector of the line whose end points are A(0, 5) and B(6, 1)
   (b) Given the vector a=6i – 2j,b= 2i +j find the scalar ''t'' and ''k'' such that ta +kb=8i-3j
5. (a)With example of diagrams briefly explain
   i. Acute angles ii. Obtuse angles iii. Reflex angle

   (b) Given the triangle ABC is similar to triangle PQR, $\overline{AB}$ = 4cm, $\overline{BC}$ = 5cm , $\overline{PQ}$=18cm and angle PQR is 30°, find the area of triangle PQR.

6. If y is inversely proportional to X^3 and Y=8 when X=2, find the value of X when Y= -1
   (b) it takes 12 days for 10 men to assemble a machine. How long could it take for 15 men to assemble the same machine?
7. A school has 3000 students, of whom 2000 are boys.
   i. What is the ratio of boys to girls in the school?
   ii. What is fraction of boys to girls .

   (b) Study the information below;

   Opening stock 01/10/2019…….shs 3,000,000/=

   Closing Stock 30/01/2019……shs 2,000,000/=

   Net purchases during 2019……shs 20,000,000/=

   Gross profit maker up…….shs 50%

   General expenses for the year……shs 4,000,000/=

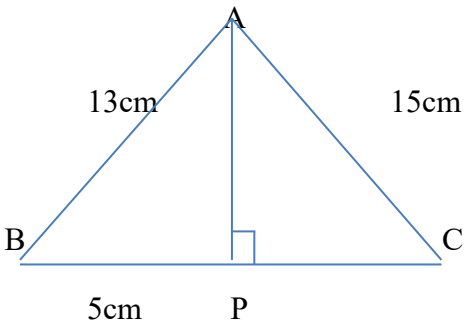
   Use the above information to find i. Costs of goods sold ii. Sales

8. (a) The first term of an arithmetic progression is 6 and the common difference is 5. Find
   i. The third term ii. The nth term

   (b) The product of first three terms of a geometric progression is 8000. If the first term is 4 .

   Find the second and third term.

9. (a) In the following figure, $\overline{AP}$ is the perpendicular to $\overline{BC}$, $\overline{AB}$=13cm, $\overline{BP}$ =5cm and $\overline{AC}$ =15cm



   Calculate the lengths of $\overline{AP}$ and $\overline{CP}$

(b) The angle of elevation of two points which are 110m apart from the top of a building are 46° and 63° respectively. Find the height of the building.

10. (a) Juma is twice as old as Ali, but John is five years younger than Ali, Find John age if their total age is 27years.
    (b) The sum of the squares of two consecutive positive numbers is 61. Find the numbers.

SECTION B (40 Marks)

11. The ages of people in certain village was recorded as follows

52	61	42	27	38	44	56	36	73	22
41	48	77	30	46	43	72	63	43	76
47	53	38	55	60	51	47	58	33	37

(a) Make frequency distribution table by grouping the ages of people in the class intervals: 20 -29, 30 -39, 40 -49, …

(b) By using frequency distribution table obtained in part (a) , calculate the mean ages

(c) Construct a pie chart for a frequency distribution obtained in part (a) .

12. find the distance between A(29° N, 72°W) and B(10°S, 72°W) in;

i. Nautical miles ii. In kilometer's

(b) (i) Find the total surface area of a rectangular prism 12cm long, 8cm wide and 6cm high.

(ii) Calculate the volume of a square pyramid whose altitude is 10cm and length of side base 6cm.

13. (a) (i) Given the matrices R=$\begin{pmatrix} 2 & -3 \\ 5 & 4 \end{pmatrix}$ and B=$\begin{pmatrix} 18 & 24 \\ -30 & 6 \end{pmatrix}$

Find 2R- $\frac{1}{6}$B

(ii) if the matrix $\begin{pmatrix} 9R & 2 \\ 8 & 4 \end{pmatrix}$ is singular, find the possible value of K.

(b) Solve the following system of linear equations by using the inverse matrix method

$$\begin{cases} 2x+3y=7 \\ 7=\frac{1}{2}x \end{cases}$$

14. (a) The function F is defined as follows

$$F(x) = \begin{cases} 1 \text{ if } x \leq 0 \\ X^2 + 1 \text{ if } 0 < x \leq 2 \\ 5 \text{ if } x \geq 2 \end{cases}$$

i. Sketch the graph of f(x)

ii. Use the graph to determine the domain and range of f(x).

(b) George buys x pencils 10 shillings each, (x+8) exercise books at 10 shillings each.

If he wishes to have some change as town bus fare. From a 200 shs note, form an inequality and solve it to find the range of x